Study on fMRI Analysis of Somatosensory Brain Activation via MR-Compatible Stimulation Device For the Standardization of Brain Scans

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Abstract: This study aims to standardize MR-compatible stimulation devices' pin and brush stimulation. The device consists of an MR-compatible stimulator placed inside the fMRI beside the participant and a tube-rod mechanism controlled outside by step motors and linear actuators. The device will stimulate the participant's palm using a brush and pin module with constant frequency and pressure. The theory behind using a device to stimulate the participant instead of a manual stimulation is to standardize the stimulus to the participant so that brain activation between scans can be predicted. The goal of using the device is to increase the contrast between the activated and non-activated areas of the brain as well as try to get a similar brain activation level between scans to contribute to giving a more objective evaluation of the neural activation associated with the stimulus, leading to a more accurate assessment to help diagnose sensory disturbances such as lesions in the peripheral nervous systems and other diseases.

1. Introduction

Functional Magnetic Resonance Imaging(fMRI) is a magnetic scanning technique that uses a strong magnetic field and radio waves to map brain activity objectively. One of the advantages of fMRI is that it is non-invasive and does not involve ionizing radiation, making it a safe and well-tolerated brain imaging method. fMRI works primarily by measuring changes in blood flow in the brain.

The fMRI scan completes a map of a participant's brain activity to help researchers and doctors better understand the region linked to critical functions such as thinking, sensing, speaking, and moving.[1] When an area of the brain is active, it requires more oxygen and glucose to function, which causes an increase in blood flow to that area. The fMRI measures this change in blood flow by detecting the difference in oxygenation. Hemoglobin is diamagnetic when oxygenated but paramagnetic when deoxygenated.[2] This difference in magnetic properties leads to differences in the MR signal of blood depending on the degree of oxygenation. This is used to identify which brain areas are active during a specific task or in response to a particular stimulus, allowing researchers to study brain function and activity.

Somatosensation is a fundamental aspect of human perception, which involves processing tactile sensations, temperature, pain, and proprioception (sense of body position and movement). By applying sensory stimuli to a participant, specific brain activation signals of regions relating to the body part can be obtained. By studying somatosensory processing, researchers can gain insights into how brains perceive and interpret physical stimuli, contributing to a deeper understanding of human cognition.

2. Background

Cotton tips, monofilaments, and tailored surfaces have been administered manually to investigate fundamental aspects of tactile perception and somatosensory processing [3]. During the fMRI scan, the participant is placed inside the scanner, and a series of scans are acquired while the somatosensory stimulus is applied. These images are then analyzed to identify which brain areas activate in response to the stimulus and the amount of activation. In traditional fMRI scans, a doctor or researcher provides stimulation manually inside the MRI room to activate the somatosensory area. However, manual stimulation can have temporal and spatial variation every time it is administered. Automating this stimulation process is an avenue to solve this problem.

However, due to the high magnetic field generated by the MRI scanner, devices containing any metallic or magnetic materials are restricted. Therefore, a specialized MR-compatible device is used in this experiment to provide consistent, measurable stimulation in an attempt to standardize the fMRI analysis of somatosensory areas activated by passive touch. Previous studies have shown that the automatic application of somatosensory stimuli during fMRI results in less variance of the BOLD time series data and

increased sensitivity for brain activations.[4] The development of this device is essential because it helps provide brush and pin stimulation to participants in a stable, reproducible way which can contribute to a better understanding of the neural basis of sensation and perception and various neurological and psychiatric conditions.

3. Materials & Method

3.1 Device Mechanism

The device consists of a stimulator module and a driver. The stimulator module is positioned within the fMRI scanner and has interchangeable brush and pin configurations. The driver consists of two linear actuators connected to a microprocessor which takes input from a PC via serial communication. The actuators drive tube-rod mechanisms that transmit mechanical power from the driver to the stimulator in the magnetic environment to stimulate the participant's fingertip, as shown in **Figure 1**.

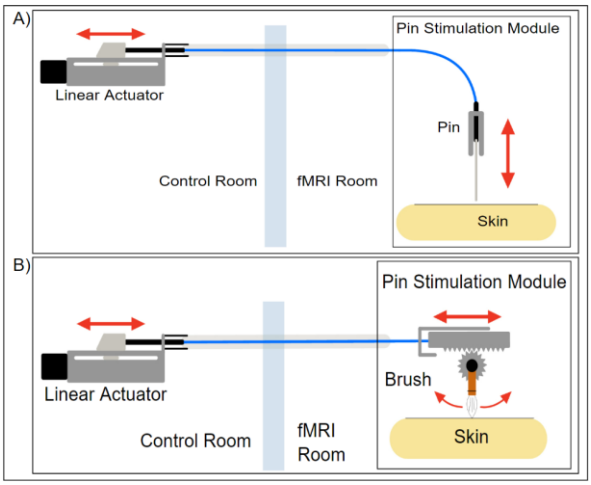


Figure 1: Pin Stimulation Module(Top)
Brush Stimulation Module(Bottom)

For the pin stimulation, the linear actuator acts to push the pin mechanism attached to the optic fiber rod into the participant's finger, as shown in **Figure 2**.

For brush stimulation, linear actuators oscillate to push and pull a pinion mechanism connected to a brush that brushes the participant's hand back and forth to stimulate the participant at a constant frequency, as shown in **Figure 3**.

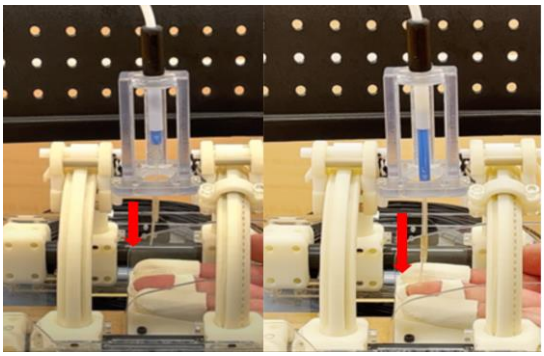


Figure 2: Pin Stimulation Module



Figure 3: Brush Stimulation Module

3.2 Method

The stimulation module is placed inside the fMRI room beside the participant, and the tubing from the linear actuator positioned outside in the control room is fed through a wall to the stimulation module positioned within the fMRI room, as shown in **Figure 4**. The device, powered by the linear drivers using a rod and tube mechanism, stimulates the participant's finger during fMRI scans to analyze the correlating brain activity.

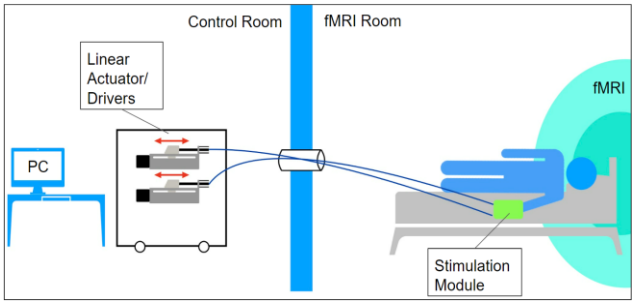


Figure 4: Device Setup in fMRI and Control Room

Scans consisting of a "Survey" and "Reference" were done to view and position where the fMRI will be scanning the brain. Then, 80 volumes of scans were taken of the brain with alternating "Rest" and "Stimulation" periods to ensure the contrast between the brain at rest and during the stimulation. Two brain scans of the device's pin and brush stimulation were captured and analyzed using MATLAB and the standard SPM12

fMRI processing software. The resulting activation areas were modeled onto a standard brain using the XJView software. The resultant images of the four scans were analyzed to ensure that the correct location in the brain correlating to the stimulation area was activated. The images were then compared regarding activation region, contrast, and quality to see if the device simulations would induce consistent brain activation patterns.

4. Result

The resulting 80 volumes of the stimulation and rest were compiled and analyzed using the SPM12, and the resulting image of the brain signals was modeled onto a standard brain for both pin and brush stimulation.

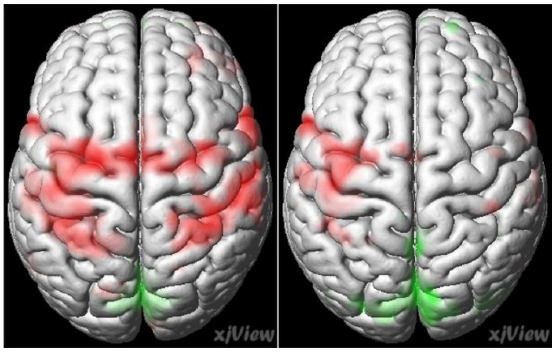


Figure 5: Participant 1 Pin Brain Activation Signal
First Scan(Left), Second Scan(Right)

The results of two scans of the same participant with the pin stimulation, shown in **Figure 5,** showed brain activation in the somatosensory area in the brain linked to the hand, as shown on the Penfield map of the brain in **Figure 6**. The Penfield Homunculus is a visual representation of the mapping of body space in the brain's somatosensory cortex, with the body's size representing the size of the area of cortex devoted to it, with the area correlating to hand sensing circled. [5]

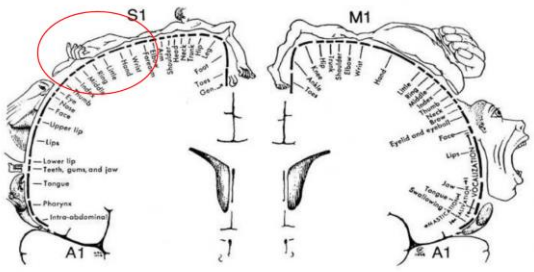


Figure 6: Penfield Brain Map

The brush stimulation scan result is shown below in **Figure 7** with the view from the top of the brain. However, other signals are getting picked up by the fMRI, which may be due to the signal from the brush stimulation being too weak.

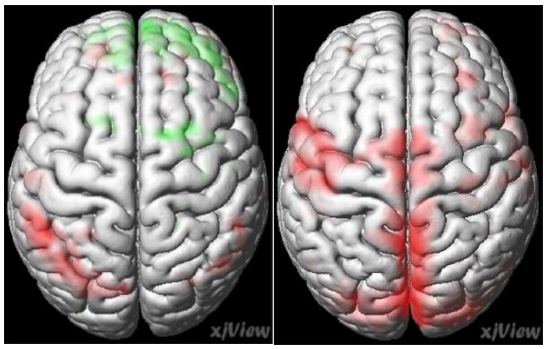


Figure 7: Participant 1 Brush Brain Activation
Scan 1(Left), Scan 2(Right)

5. Discussion

When we compare the signals from the pin and brush, the pin stimulation signal contrast and location are more apparent than the brush stimulation. The signal from the brush stimulation predominantly activates mechanoreceptors responsible for light touch, which might be lost in other sensory inputs such as the pressure from the fMRI bed, clothes, light, and noise, which can pick up signals in the image unrelated to the stimulation, making it difficult to interpret the results accurately. In contrast, pin stimulation allows for the focused application of pressure to the finger, resulting in highly localized stimuli, which is more likely to produce a distinct and differentiable sensation, making the pin a better stimulant than the brush. However, the pin stimulation also comes with its caveats. While the pin has a more localized signal than the brush, the pin stimulation can activate the nociceptive (pain) receptors, as after the first pin scan, Participant 1 reported that the prick from the pin made their face wince in response. This would explain the signal in the brain's right hemisphere, as shown in the first scan in **Figure 5**, which correlates to facial movement. However, after the initial scan, the participant did not wince, which can explain the disappearance of the signal in the second scan shown in **Figure 5**. While the signal between the scans of the same participant yielded a slightly different activation, the pin stimulation showed a good correlation between inter-participant scans.

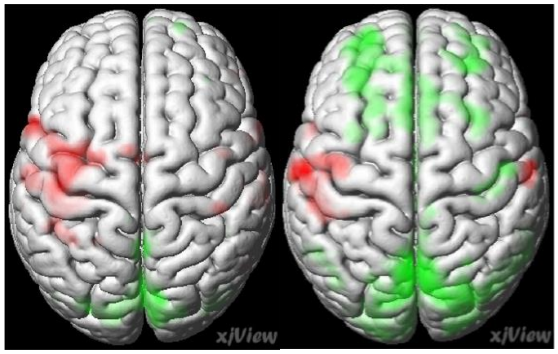


Figure 8: Pin Stimulation Brain Activation Signal Between 2 Different Participants
Participant 1(Left), Participant 2(Right)

When using the pin stimulation module on two different participants, as shown in **Figure 8**, the signals showed that the somatosensory-activated area and intensity stayed relatively the same, which shows that this technique has improved the standardization across different participants.

6. Conclusion

This experiment aimed to reduce the variability in image quality and the acquired data, increasing the sensitivity of the scans to detect smaller effects to improve the accuracy and clarity of the findings. There are several issues associated with fMRI imaging, such as being easily influenced by noise sources such as head movements and breathing, making the standardization of brain activity signals of fMRI scans challenging. While the aim was to standardize the stimulation technique of multiple scans with different participants, leading to more reliable and valid results, many variables and considerations must still be addressed to accomplish standardization.

Both brush and pin stimulation techniques have their merits and can provide valuable insights into sensory processing. However, pin stimulation offers certain advantages regarding precision, discrimination, intensity control, experimental flexibility, and targeting specific sensory subtypes. However, as shown in this experiment, the pressure applied significantly affects the signal acquired. With stimulation to a painful degree, other areas in the brain may be stimulated by reflex in response to the stimuli. Also, the variability with the individual subjects being scanned makes it challenging to standardize images. Different individuals can have different head shapes and sizes, leading to variations in image quality and brain activity measurements. However, using a device to induce the same stimulation every time can improve image quality and data preprocessing variability, leading to more reliable and valid results that impact the medical field and statistics, computational neuroscience, computer vision, and computer science.

With the advent of AI's recent incorporation into medical studies to diagnose diseases based on medical images using fMRI, the standardization of the stimulation can help improve the statistical power of these scans. By reducing variability in the data, standardized scans can increase the sensitivity of each scan, allowing for the detection of smaller effects and changes to brain activity, leading to better and faster diagnoses of neurological conditions.

References

1) Tung, M. (n.d.). What is an fMRI scan, and how does it work? https://www.jameco.com/Jameco/workshop/Howitworks/what-is-an-fmri-scan-and-how-does-it-work.html

2) Functional_magnetic_resonance_imaging. (n.d.). Functional_Magnetic_Resonance_Imaging. https://www.bionity.com/en/encyclopedia/Functional_magnetic_resonance_imaging.html

3) Davis et al., 1998; Fox et al., 1987; Francis et al., 2000; Hagen and Pardo, 2002; Iannetti et al., 2003; Moore et al., 2000; Ruben et al., 2001; Yetkin et al., 1995

4) Murata K, Matsushita A, Saotome K, Kawamoto H, Sankai Y. Development of an MR-compatible configurable brush stimulation device. Annu Int Conf IEEE Eng Med Biol Soc. 2014;2014:2101-6. doi: 10.1109/EMBC.2014.6944031. PMID: 25570399.

5) Proulx MJ, Todorov OS, Taylor Aiken A and de Sousa AA (2016) Where am I? Who am I? The Relation Between Spatial Cognition, Social Cognition and Individual Differences in the Built Environment. Front. Psychol. 7:64. doi: 10.3389/fpsyg.2016.0006

Appendix

fMRI User Manual

1. Participant Loading

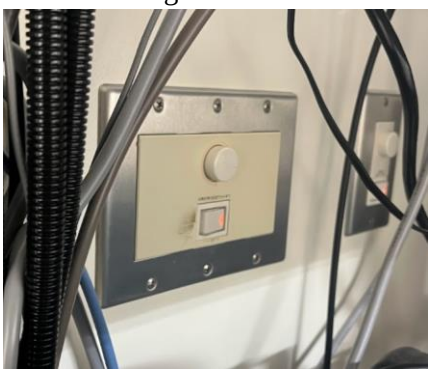
1.1 Before the Scan:

1. Use the MRI screening form to question the patient to ensure no metal fragments or neurological issues(claustrophobia, etc.)
    - The form is in the topmost drawer of the white cabinet

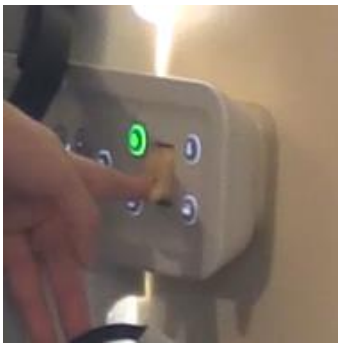


2. Make sure that the participants do not have tattoos
3. Remove any metal or electronics on the patient and yourself. (Triple Check)
    - If the person needs glasses, have them wear the med-goggles for fMRI located in the black box in the lockers.
    - Put all the belongings in the lockers.
4. Change clothes into the surgical gown in the locker
5. Explain the exercise that will be taking place ahead of time to the patient because it is loud inside the MRI room.

1.2 Loading the Participant in fMRI Machine

1. Turn on the light in the fMRI room at the back of the shelf
2. 

3. Pull the table out and lower it down by pressing down the center switch in the control panel



4. Take out a towel from the second drawer of the white cabinet inside the MRI room and put it down where the head will rest




5. Connect the Head Coil to the bed socket if left unplugged.



6. Explain the task to the patient and lay the patient down on the fMRI bed
    - Practice the task with the patient and make sure the patient knows what they have to do (ex. Leg moving, hand tapping)

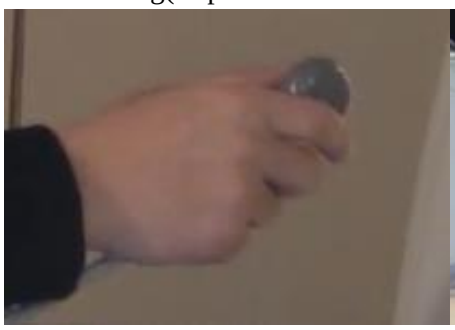
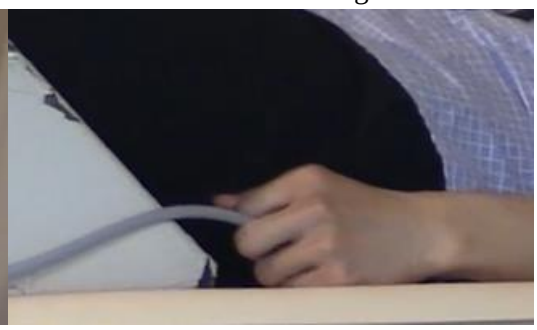
7. Put earplugs on the patient and put on the headphones




8. Put the pillow under the legs



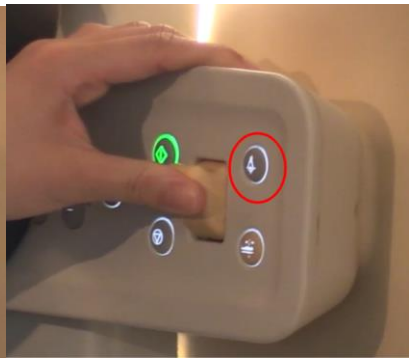
9. Plug in the buzzer to the plug near the legs and put the buzzer in the hand that they will not use while scanning(Explain how the buzzer works and how to use it to signal to the PC room)

10. Put the headgear(light visor) on the patient’s head and adjust the light visor so that it is directly over the patient’s eyes



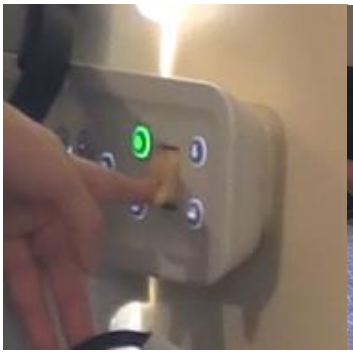

11. Lift the bed and press the top right button to turn on the laser. Adjust the bed's position to center the laser crosshairs on the visor.

12. Press the bottom right button to send the patient into the scanner



13. Ask the patient if they can see the projection
    - If not centered, pull them out using the switch and adjust the visor and bed position as needed.




- Long press the bottom right button to send them back into the scanner




2. fMRI Operator/Slides Operator

2.1 The setup of the Control room:

1. Turn on the PC labeled PC 1, **刺激制御用コンピューター**



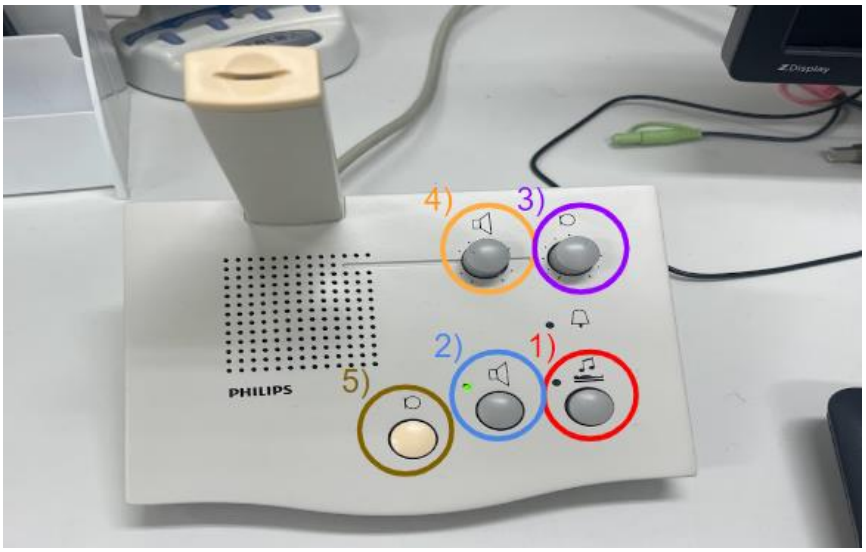
2. Make sure that PC 1 is selected on the monitor switchboard



3. Connect your PC to the PC 1 monitor via a VGA cable and start the slideshow



2.2 Using the PHILLIPS mic:

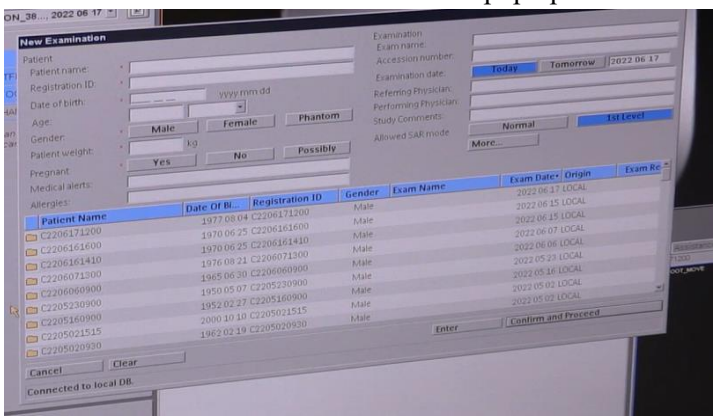


1) Press the lower right button to activate the intercom so that the person inside the fMRI can hear
    a) Make sure the lower right button is green. If it is off, the patient cannot hear you.
2) Press the lower left button to activate the speaker on the mic
    a) Make sure that the lower left button is green. If it is off, you cannot hear the patient
3) Adjust the volume for the person inside the fMRI using the top right knob
    a) Speak with the patient so that the volume is comfortable for them
4) Adjust the volume for you using the top left button.
5) Press the center red button to talk to the person inside the fMRI machine.
    - Remember that it is a one-way mic, so to hear their reply, take your finger off the red talk button.

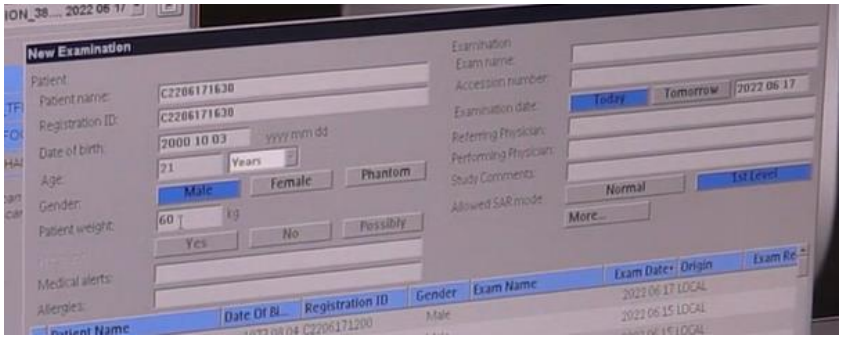
2.3 Using the fMRI:

1. When on the Philips Application Login screen,
   The username is ‘mruser,’ and leave the password blank.

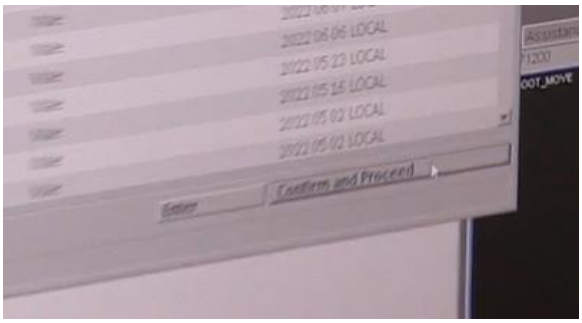
2. Once logged in, Press F2
    - The ‘New Examination’ window should pop up



3. Insert Information into all boxes that have an asterisk (Do not use the real names or date of birth of the patient as this might result in a breach of privacy)
    - Patient name
        - C+YYMMDDTTTT (ex: C2305191730)
    - Registration ID
        - Copy the Patient’s name and paste
    - Date of Birth
        - Put your Date of Birth so that you can see that you took the scan when reviewing
    - Age
        - If you press ‘Tab’ after inputting the date of birth, it will fill it out for you automatically
    - Gender
        - Male (doesn’t matter)
    - Patient Weight
        - Put in the weight of the patient. This will determine the amount of electromagnetic waves the patient is exposed to. (Add more weight just to be safe)

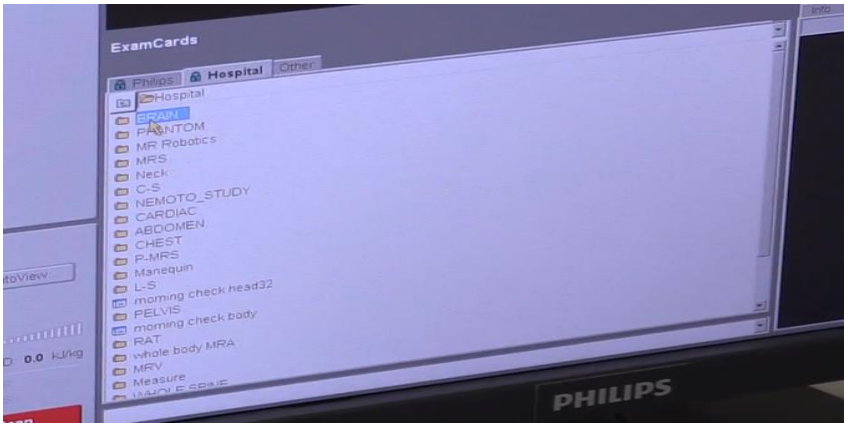


4. Hit ‘Confirm and Proceed’

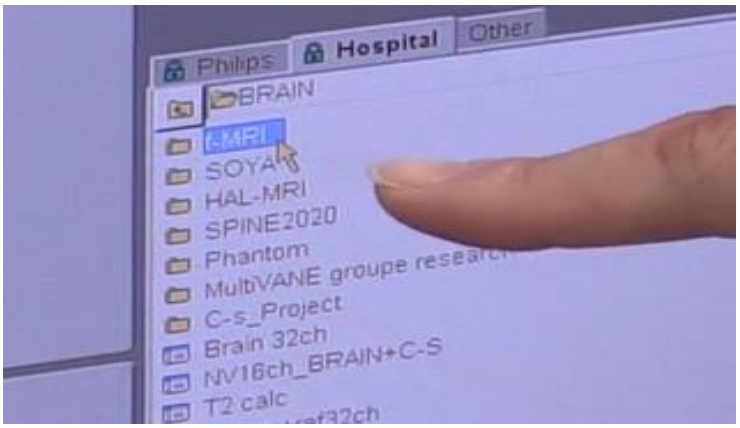


5. Go to the Exam Cards with the Hospital Tab on the bottom center

6. Select the BRAIN file



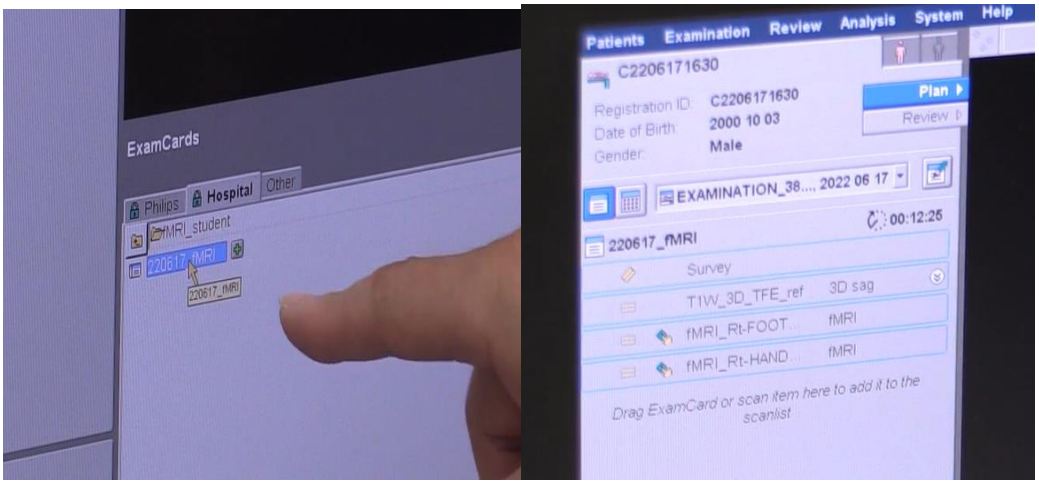
7. Select the fMRI file



8. Select the fMRI_student file (Scroll to the very bottom)

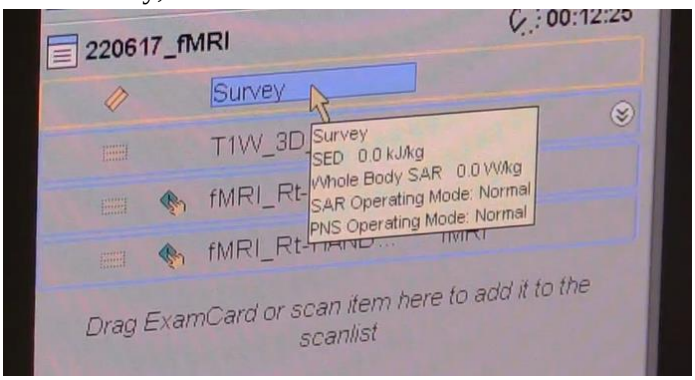


9. Highlight the YYMMDD_fMRI and drag and drop it into the left tab



- Survey, T1W_3D_TFE_ref, fMRI_Rt-FOOT, and Rt-Hand should pop up
    - Survey: Used for testing to get the image
    - T1W_3D_TFE_ref: Used as a position reference to increase image accuracy

10. Click Survey, and the box around it should turn from blue to orange (this means that it is active)



11. Tell the patient over the mic that the scan will start and that there will be loud noises coming from the fMRI

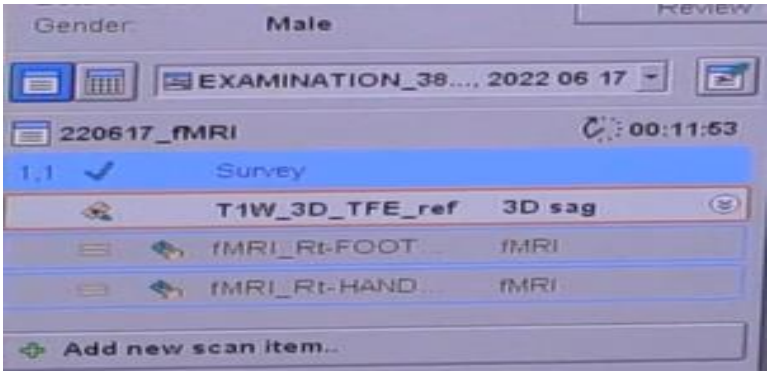
12. Press Start scan



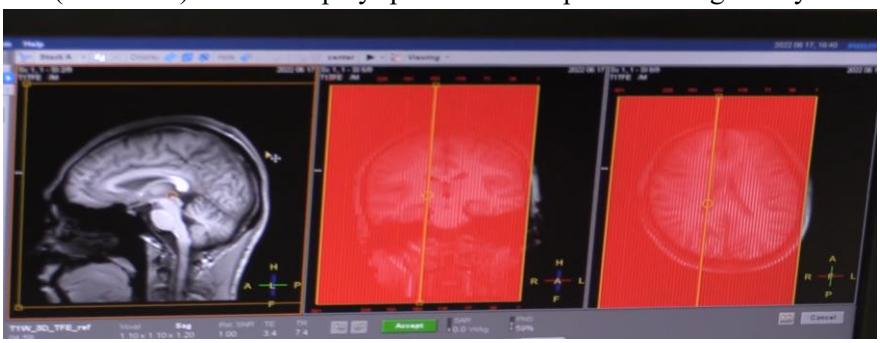
13. Wait for about 5 minutes until the fMRI images come up

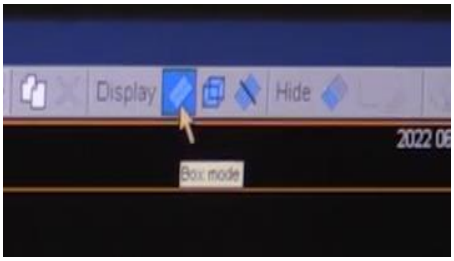


14. Double-click the T1W tab to make the T1 active(it should have an orange box around the perimeter)



15. If a red box comes up on the screen, making it hard to see, click the right(all mid planes) and left icon(box mode) of the ‘Display’ panel on the top to make it go away







16. Click on only the orange and blue boxes to change the scanning region(Make sure not to touch the left image. Only manipulate the center and right image)
    - Click and drag the orange circle in the middle to move the scanning region



    - Click and drag the top orange box to change the angle of the scanning region

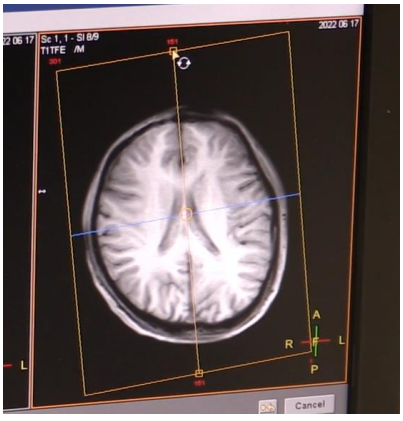


- (do not touch the lines, as they can jump to different images and change the page numbers)

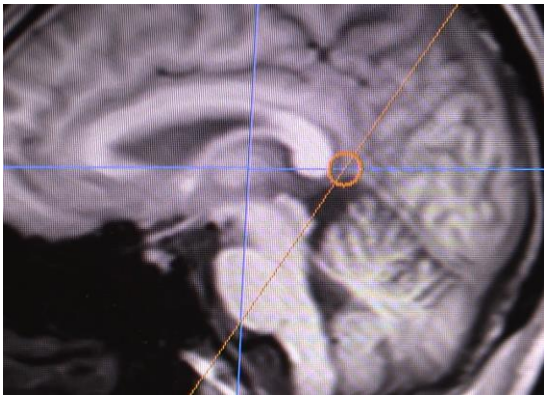
17. Middle image: Line the orange circle with the center of the brain and the orange line with the brain center line



18. Right image: Tilt the angle so that the orange line aligns with the brain center line



19. Left image: Make sure the blue cross is centered at the optic thalamus using the orange circle

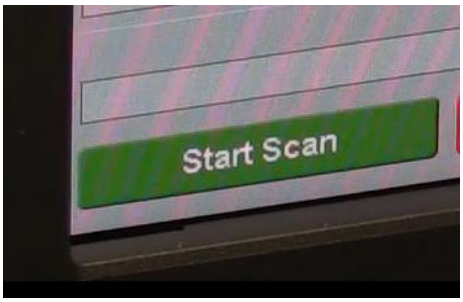


20. Press the green ‘accept’ button



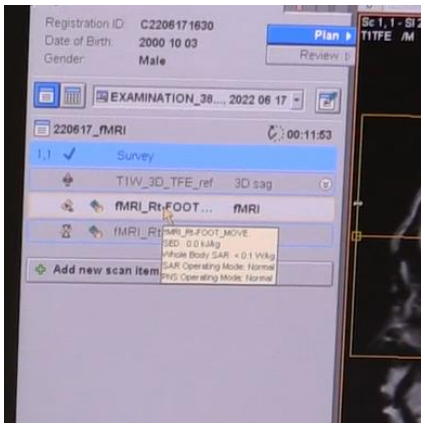
21. Tell the patient over the mic that the scan will start and that there will be loud noises from the fMRI for about 5 minutes

22. Press ‘start scan.’

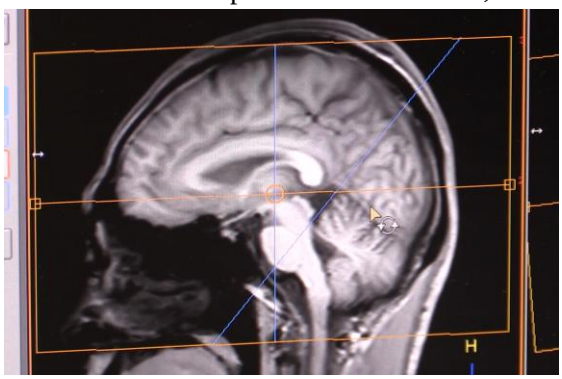


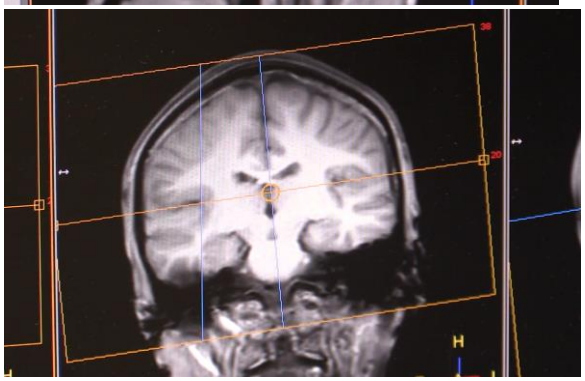
Prepare the following scan: While we wait for about five minutes for the image to be captured, we can start planning for the next scan

23. Double-click on the fRMI_Rt-foot on the left panel



24. Use the orange center circle to adjust the left image so that the orange line bisects the anterior commissure and the posterior commissure, and align the angle of the left image





Adjust to leave some room in the top, even if it means the bottom gets cut out because we are trying to see the brain activation at the top of the head for this scan

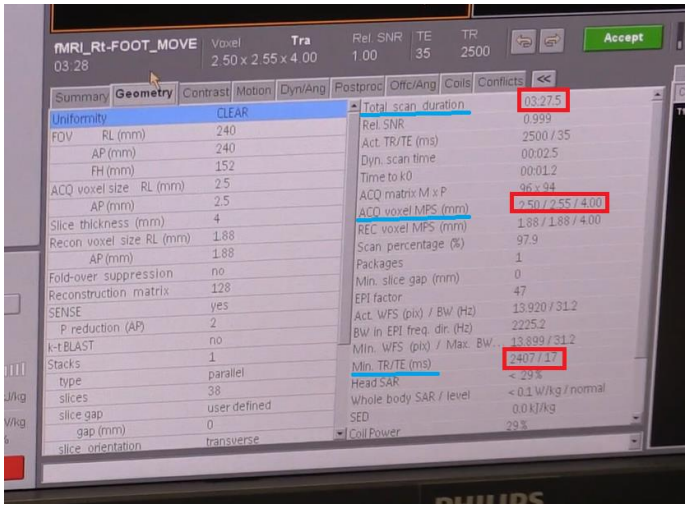
25. Press accept



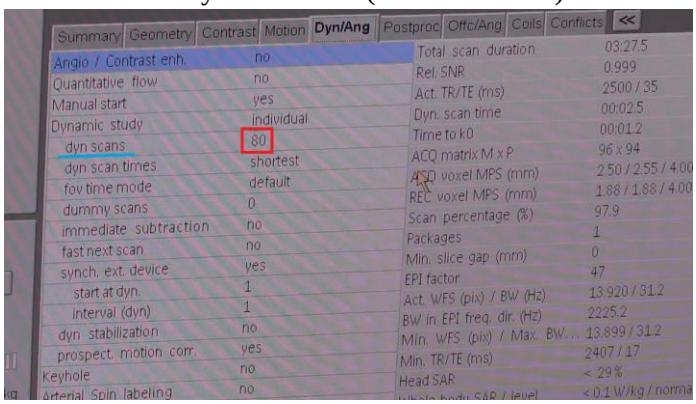
Information for Analysis:

Total Scan Duration, TR/TE(ms), ACQ voxel MPS(mm)

- Click geometry
  - Total Scan Duration
  - TR/TE(ms)　2500/35 40 ← voxel size
  - ACQ voxel MPS(mm)　←



- Click Dyn/Ang
  - Dynamic scans (80: 10 sets of 8)



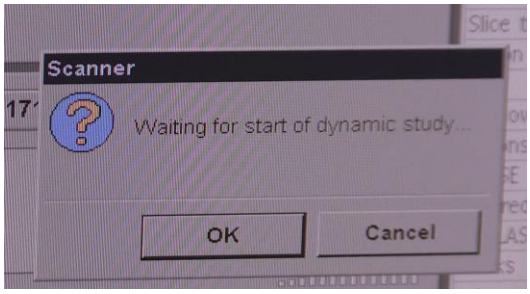
26. Make sure that the first scan is finished and press ‘Start Scan’ for the Rt_Foot scan



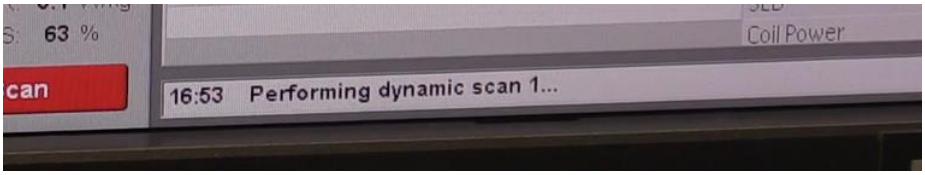
    - The scan should not start right away; it is going into prescan, which checks how much EMW goes out and back in

27. Ensure the slide show shows the ‘X’ slide for the rest period (1-10)

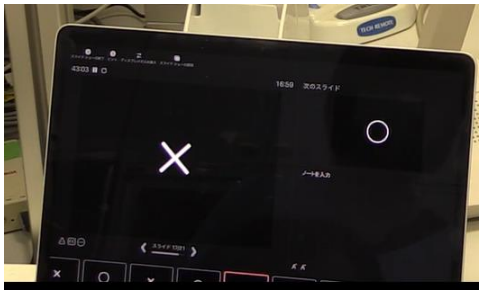
28. A ‘Scanner tab’ should pop up; click ‘ok’ to start fMRI
    - Make sure to tell the patient that there will be a loud noise so that they will not be surprised



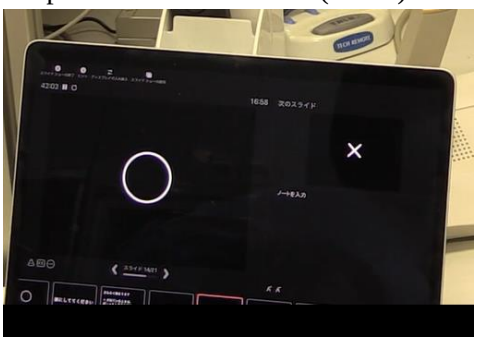
    - “Performing dynamic scan #” should pop up at the bottom. This means that the fMRI is being performed.



29. The first 1-10 dynamic scans should be for the ‘rest’ period, meaning the ‘X’ should be shown on the slideshow screen.



30. Right when the 11th dynamic scan comes up, you should switch the slides to the ‘O’ signaling to the patient to start the task (11-20)



    - And we alternate until the 80th scan is complete

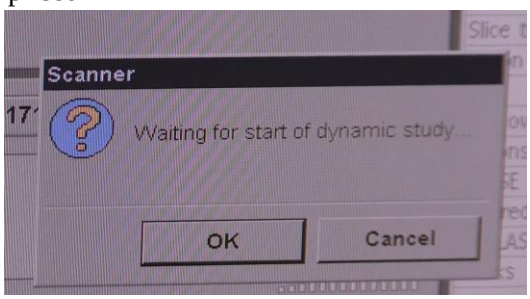
31. Make sure that the first scan is finished and press ‘Start Scan’ for the Rt_Hand scan



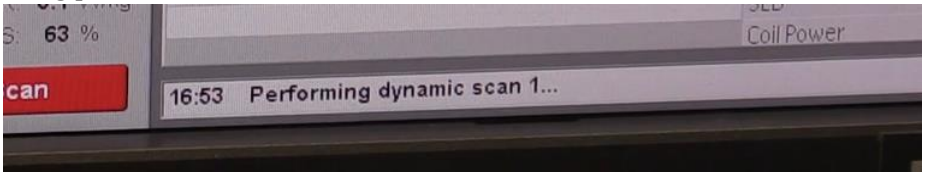
    - The scan should not start right away; it is going into prescan, which checks how much EMW goes out and back in

32. Ensure the slide show shows the ‘X’ slide for rest period (1-10)

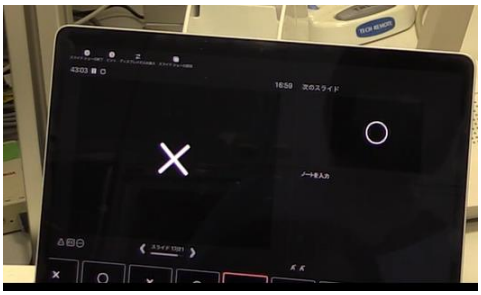
33. A ‘Scanner tab’ should pop up; click ‘ok’ to start fMRI
    - Make sure to tell the patient that there will be a loud noise so that they will not be surprised



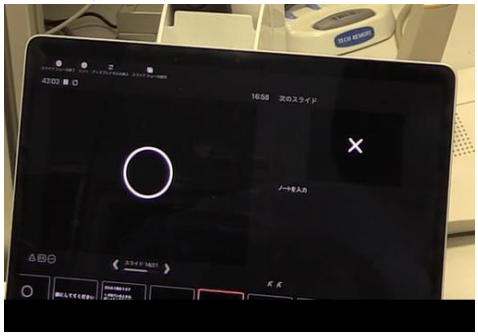
    - “Performing dynamic scan #” should pop up at the bottom. This means that the fMRI is being performed.



34. The first 1-10 dynamic scans should be for the ‘rest’ period, meaning the ‘X’ should be shown on the slideshow screen.



35. When the 11th dynamic scan comes up, you should switch the slides to the ‘O’ signaling to the patient to start the task (11-20)



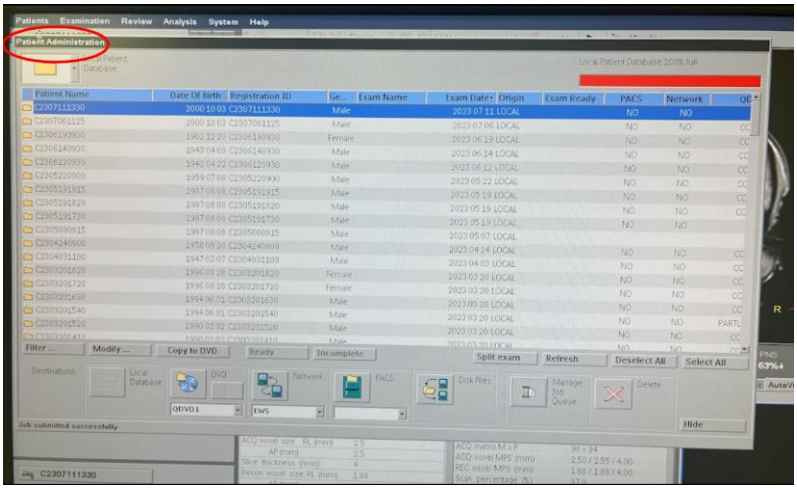
36. And we alternate until the 80th scan is complete

2.3 Export Images

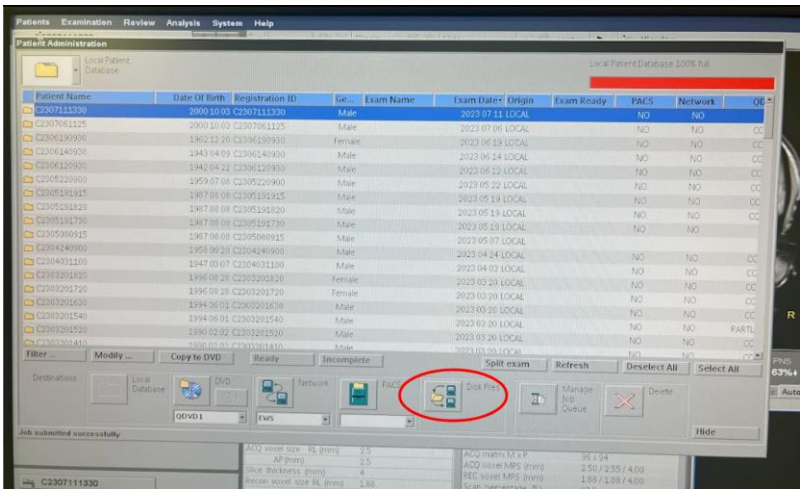
1. Insert a USB thumb drive into the Control PC tower
2. Click on the Examination Tab at the top left



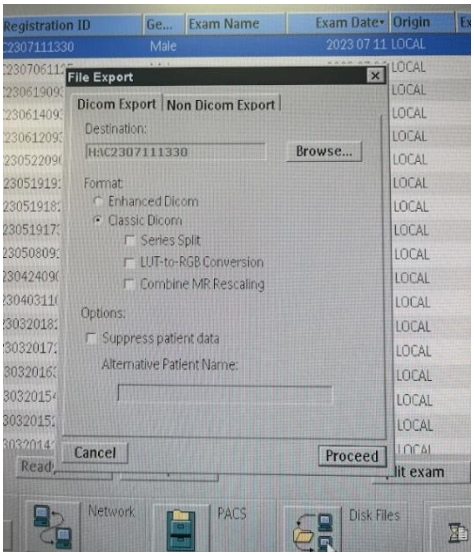
3. Click on the “Administration”, and a “Patient Administration” tab should pop up



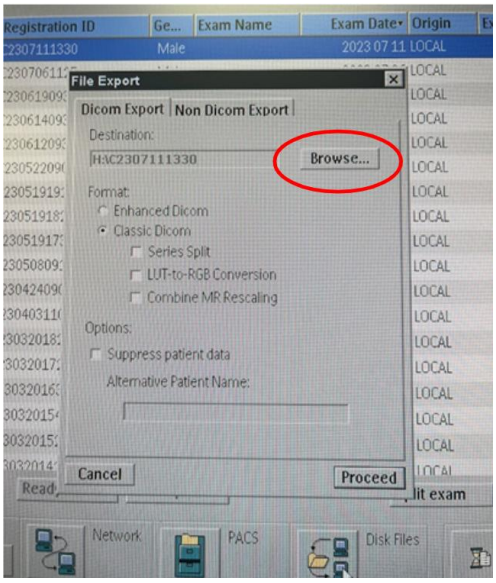
4. Click on the “Disk Files” icon



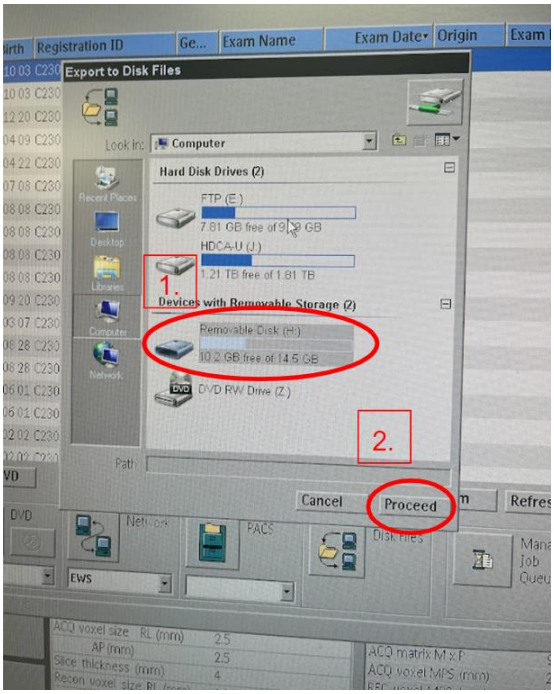
- A “File Export” tab should pop up.



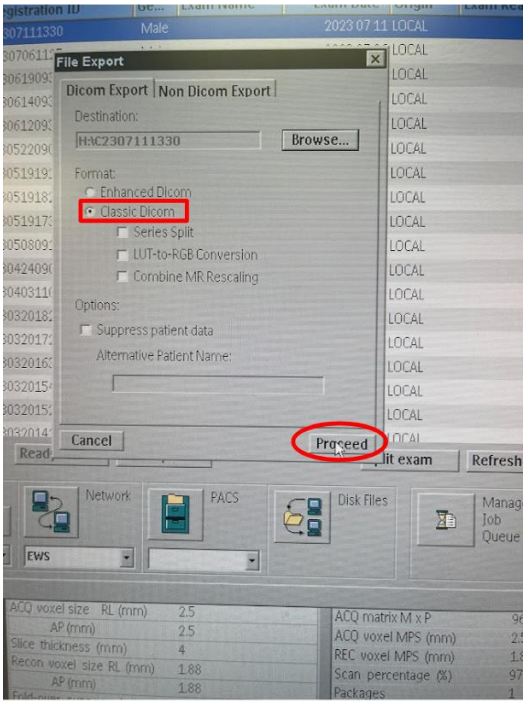
5. Click on the “Browse…” button



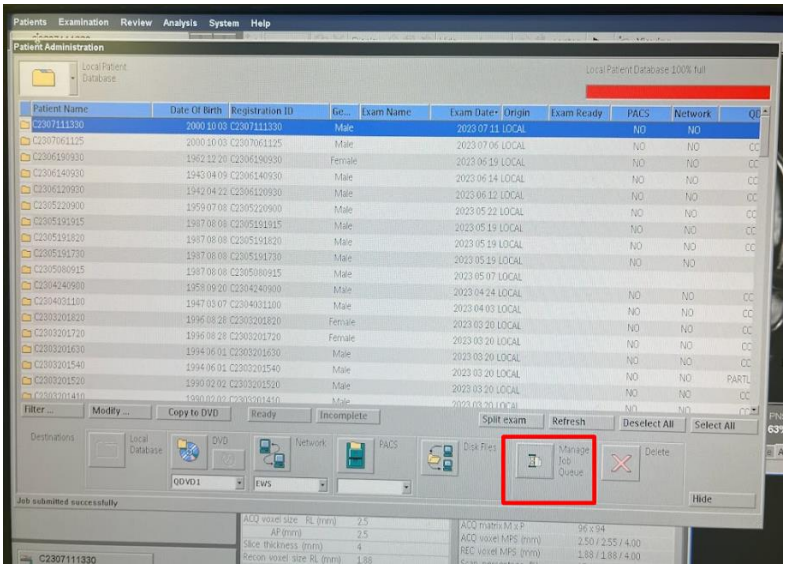
- An “Export to Disk Files” tab should pop up. Select the Removable USB that you want to send the DICOM files to and press “Proceed”



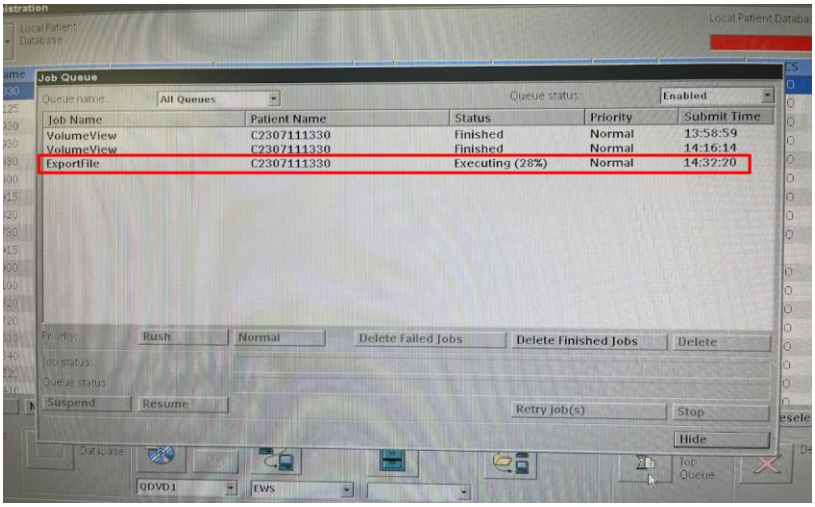
- Once you hit “Proceed,” the screen should return to the “File Export” Tab.
  - Check that the “Classic Dicom” format is selected and hit “Proceed”
  - DO NOT CHECK THE OPTIONS BELOW THE CLASSIC DICOM, OR THE CONVERSION PROGRAM WILL NOT WORK



6. To check the progress of the export, Click “Manage Job Queue” to see how the export is going



- A Job Queue tab should pop up
  - Once it reaches 100%, pull out the USB drive and check on your own laptop that the Dicom files are on the USB drive.



Manual for Analysis of the fMRI Data

*Instructions are for macOS

The Prerequisite programs that need to be installed

- Python programming language
  Necessary to run FSL scripts. (Installed by default on mac OS)
- XCode (Apple's integrated development environment)
  - (https://softantenna.com/blog/download-xcode/) for your Mac version
  - Required to compile SPM12 which is a large file (about 10GB)
  Install ahead of time because it takes a long time
- MATLAB (Matrix arithmetic software)

https://www.cc.tsukuba.ac.jp/wp/service/sl/matlab/

  - Install MATLAB from the above (Installing all toolboxes recommended for future use)
- MRIcron (For DICOM to NIfTI file conversion)
  - Go to the Nitric site and download the newest version

https://people.cas.sc.edu/rorden/mricron/dcm2nii.html

1. Install and extract mricron_macOS.dmg from the above URL
2. Confirm that MRIcron can be opened

- SPM12 (Statistical image analysis package)
  https://www.fil.ion.ucl.ac.uk/spm/software/download/
  1. Select PC OS and MATLAB from the above and install zip
  2. Extract the two zip files (mricron_macOS.dmg)
     https://en.wikibooks.org/wiki/SPM#Installation
     a. The path is optional, but in this case, unzip spm12.zip under /Users/Uehara/Documents/MATLAB
  3. Start MATLAB and add the path to SPM on the command line (see URL above)
     >> addpath /Users/Uehara/Documents/MATLAB/spm12
  >> savepath
  4. Compile with Xcode (see URL above)
  5. On the MATLAB command line, type >> edit spm_defaults.m to open the SPM defaults file in the MATLAB editor and change defaults.images.format = ‘img’ ; %~~ img to nii, and then press command + s to overwrite the file.
  6. confirm that you can start MATLAB by typing >> SPM on the command line
- FSL (a comprehensive library of analysis tools for fMRI, MRI, and DTI brain imaging data)
  https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FslInstallation
  1. Download and run the python script for your OS from above.
  2. Confirm that it can be started by typing "fsl" in the terminal

For Analysis

You can put the data where you want, but putting it inside the MatLab file is recommended.

- Create a file name called 2022CYBfMRI within the MATLAB file and create two subfiles within the 2022CYBfMRI file called ‘nii’ and ‘Original.’
- The ‘Original’ file should contain the DICOM data from the fMRI terminal.
- Inside the DICOM file, there should be files labeled 00000001, 00000002, and 00000003, as well as executable files called IM_XXXX. The PHILIPS device can only export the data in the directory in 2050 file groups. The data was exported in these 0000000X files with the remaining IM_XXXX files on the bottom.
    - The data is exported differently for every device, so keep that in mind.
    - The IM_XXXX files are labeled as executable files but are DICOM files
    - We will be converting these files into nii files
- Get the downloaded MRIcroGL and put the file into the applications
- Download the template.mat file
- Open a new terminal window and go to the directory that contains the MRIchroGL.
    - Go to the terminal window and type ‘cd,’ drag and drop the MRIchroGL file after the cd, or type ‘/Applications/MRIcroGL’ and hit enter to move to that file.
    - Type in ‘ls’ to ensure you are in the right file. You should get a list of the files within the MRIcroGL file.
    - You should see a program called dcm2niix* within that list. This is a program to convert Dicom files into Nii files.
    - Type ‘./dcm2niix*’ and then go into the C20226171700 and locate the DICOM file within. Drag and drop that DICOM file into the terminal window after the ‘./dcm2niix*’ and hit enter.
    - You should then see files created within the DICOM folder labeled ‘.nii’, also known as a NifTI file which is a convenient file for neuroimaging analysis. The file should include the name of what the fMRI was of(in this case, it was ‘Rt-foot-move’, which means that it is the image of the brain while moving the right foot). The image specifications are as follows: 38 images at 4mm intervals 80 times.
    - There should also be .json files with the same name as the nii files. These files are the conditions when taking the images and not the images itself.
- Option 1: Open the nii files in fsleyes (EASY)
    - Ensure you are in the Applications/MRIcroGL$ folder in the terminal and type in ‘fsleyes’ and hit enter. The fsleyes window should come up with a black screen in the middle. Drag and drop the nii file that you wish to see into the black screen and the brain image should come up.
    - There are 80 volumes that you can look through using the volume up and down arrows. However, the first volume is 0, so the 80th volume will be 79.
    - It is important to look through all the volumes one by one to ensure that there are no warped images. Warped images might occur because the patient moves or if there is an outside signal from big machines turning on, ambulances passing by, etc. It is imperative to take out the data of the warped volume image before analysis or the data might be affected negatively.

    - Create a folder called ‘C2206171700’ in the nii folder
    - Copy the 4 .nii files from the DICOM folder and paste them into the C2206171700 folder.

- Once done copying, create 3 more folders within the C2206171700 folder called:
  - Rt-foot
  - Rt-hand
  - T1
- Once you made these files, drag and drop the 4 .nii files into the respective folders.
  - DICOM_fMRI_XXXXRt_Foot_MoveXXXX.nii ⇨ Rt-foot folder
  - DICOM_fMRI_XXXXRt_Hand_MoveXXXX.nii ⇨ Rt-Hand folder
  - DICOM_T1W_3D_TFE_ref_XXXXXXXX.nii ⇨ T1
- Copy and paste the given template.mat files into both the Rt-foot and Rt-hand folders.
- You can delete the files labeled DICOM_Survey_XXXXXX.XXX because they are simply test runs done by the machine at the beginning.
- The JSON files are the standard of mri imagining worldwide, with the same format.
- The DICOM_VT1W_3D_XXXXXX is the image taken from the fMRI machine without any formatting. (You can delete this file, as we will not be using this for processing.
- The DICOM_T1W_3D_XXXXXX is the fMRI image that is reformatted to align with the AC-PC lines we established during the fMRI operation planning step, which should line up with the anterior commissure and posterior commissure.
- We will then use the aligned DICOM_T1W_3D_XXXXXX and merge the blurry nii files to assemble a clean image.

- Open Matlab
  - Go to the command line and type in ‘spm fmri’
  - Click on ‘Batch’
  - Click on the Load Job File button.
    - Your directory should be Users/XXXX/Documents/MATLAB
    - In this window, clicking on the folders in the left panel will go into the folders, while clicking on the files in the right panel will select the files.
  - Click on ‘2022CYBfMRI’ on the left panel, then ‘nii’, ‘C22XXXXXX’, and ‘Rt-foot’
  - Then ‘template.mat’ should show up on the right panel, click it and it should disappear and the selected file should show up on the bottom panel. Click done.
  - Job Program Selection
    - 1. Named Directory Selector:
      - Make sure that the input name is set on currency
      - Double-Click ‘. Directory’ and click on ‘2022CYBfMRI’ then ‘nii’ in the left panel. Select the ‘C22XXXXXX’ file on the right panel and press done.
    - 2. Named Directoy Selector:
      - Double-Click ‘. Directory’ and click on ‘2022CYBfMRI’, ‘nii’, ‘C22XXXXXX’. Select the ‘Rt-foot’ file on the right panel and press done.
    - 3. Named Directory Selector:
      - Double-Click ‘. Directory’ and click on ‘2022CYBfMRI’, ‘nii’, ‘C22XXXXXX’. Select the ‘T1’ file on the right panel and press done.

- Click Save Batch
    - Click on ‘template.mat’ and change the name to ‘rt-foot.mat’ and press Save.
- Delete ‘File Set Split’
    - Right-click and press ‘delete module’.
- 4. File Selector (Batch mode)
    - Double Click ‘Filter’ and change ‘^s[1-9]*’ to ‘^DICOM.*’
- 5. File Selector (Batch mode)
    - Double Click ‘Filter’ and change ‘^s[1-9]*’ to ‘^DICOM.*’
- 6. Realign: Estimate and Reslice
    - Click on ‘Session’ and click the ‘Dependancy’ button.
    - Click on the First ‘File Selector (Batch Mode): Selected Files (^DICOM.*)’
    - Click ‘OK’

○ Click Save Job

- Click on ‘rt-foot.mat’ and press save. It is ok to overwrite.
- Click Run
    - You might need admin access
- The processing functions if you see the red bar and the graphs in the background.
- You should wait a couple of minutes for the program to finish.
- Once it is done, you can use the UI to manipulate the data however you wish